Figure 6.2 Kaan (TF-X) – 2023: Turkey's fifth-generation stealth multirole fighter.⏎

Technologically, Kaan incorporates AI-enabled mission systems, sensor fusion, and advanced avionics comparable to those of the F-35 Lightning II. It features active electronically scanned array (AESA) radar, electronic warfare (EW) systems, and 360-degree situational awareness sensors, all integrated through real-time data fusion. These capabilities allow the aircraft to perform autonomous threat detection, target prioritization, and data sharing with allied platforms under network-centric warfare (NCW) frameworks.

One of the defining characteristics of the Kaan program is its plan for manned and unmanned interoperability. Future variants are expected to include an unmanned combat air vehicle (UCAV) configuration, capable of functioning as a loyal wingman in coordination with Turkey’s Anka-3 and Akinci drones. This ensures flexible mission execution and force multiplication through AI-assisted formation flying and cooperative targeting.

Powered by twin turbofan engines (currently GE F110s, with indigenous engines under development by TRMotor), the aircraft is expected to achieve speeds exceeding Mach 2, an operational ceiling of 55,000 feet, and a combat radius of 1100 kilometers. These features make it a formidable asset for both defensive air control and offensive deep-strike missions.

Strategically, Kaan (TF-X) symbolizes Turkey’s technological maturity and geopolitical aspiration to become a global defense leader. It not only strengthens Turkey’s national defense autonomy but also enhances its export potential in the competitive next-generation aircraft market.

6.5.2.3 KF-21 Boramae – 2021 (Block III Planned)

The KF-21 Boramae is South Korea’s indigenous multirole fighter air-craft, developed by Korea Aerospace Industries (KAI) in collaboration with